# Aaryan Kulkarni

Georgia Tech Electrical Engineering | AI, Hardware, RTL, Computer Vision | akulkarni456@gatech.edu | 470-586-2849

Building intelligent hardware and real-time engineering systems across AI, embedded design, and semiconductor technology.

## Education

**Georgia Institute of Technology** - B.S. Electrical Engineering, GPA: 4.0

## Experience

**Cloud Supply Chain Solutions - Software Engineering Intern**
Built real-time AI driver safety monitoring with Python, OpenCV, live video processing, dispatcher dashboard features, fatigue-risk detection, alerts, camera metadata, clickable event snapshots, and Ollama-powered local AI summaries.

**Georgia Tech Compound Semiconductor Products - Incoming Undergraduate Research Assistant**
Supporting compound semiconductor product development through characterization, electrical testing, materials and device physics, lab documentation, and technical reporting.

**Vizun - Co-Founder**
Co-founded an AR interior visualization startup with 20+ materials and layouts, market validation, financial modeling, $2,700+ seed funding, and 1st place at TiE Atlanta.

## Projects

**EvoFabric - Self-Adaptive RISC-V FPGA Compute Fabric**
SystemVerilog, RISC-V, FPGA, Verification, GTKWave
Simulation-first RISC-V-style FPGA fabric with MMIO accelerator registers, runtime selection, safety fallback logic, performance counters, Popcount, CRC32, FIR, matrix-multiply accelerators, automated testbenches, and fault injection.

**TinyRV32 - Modular RV32I Single-Cycle CPU Core**
SystemVerilog, RISC-V, Digital Design, Icarus Verilog, GTKWave
Modular RV32I CPU core with PC logic, instruction memory, ALU datapath, register file, branch control, decode/control logic, and waveform-based verification.

**Scooter Distance Warning System**
Arduino, C++, HC-SR04
Scooter-mounted obstacle detection prototype with ultrasonic sensing, adaptive buzzer alerts, 2-400 cm distance measurement, and 25+ proximity tests.

**Monte Carlo Engineering Simulation Lab**
Python, NumPy, SciPy, Matplotlib
Monte Carlo simulations for numerical integration, turbine blade reliability, projectile variation, failure probability, convergence behavior, and risk visualization.

## Technical Skills

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| **Hardware / RTL** | SystemVerilog, Verilog, RISC-V, FPGA architecture, digital logic, FSMs, MMIO, testbenches, VCD waveform debug, Icarus Verilog, GTKWave |
| **AI / Software** | Python, C++, OpenCV, real-time video processing, object detection, NumPy, SciPy, Matplotlib, Ollama |
| **Embedded / Product** | Arduino Uno, HC-SR04, sensor integration, circuit debugging, Git/GitHub, VS Code, Arduino IDE |
| **Engineering / Business** | Market research, financial modeling, startup strategy, product validation, technical reporting |

## Direction

Pursuing internships, research, and projects in AI hardware, semiconductor technology, FPGA/RTL design, embedded systems, computer vision, and safety-critical engineering.